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CNN-Transformer and Channel-Spatial Attention based network for hyperspectral image classification with few samples

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ABSTRACT

Hyperspectral image classification is an important foundational technology in the field of Earth observation and remote sensing. In recent years, deep learning has achieved a series of remarkable achievements in this area. These deep learning-based hyperspectral image classifications typically require a large number of annotated samples to train the models. However, obtaining a large number of accurate annotated hyperspectral images for high-altitude or remote areas is usually extremely difficult. In this paper, we propose a novel algorithm, CTA-net, for hyperspectral classification with a small number of samples. First, we proposed a sample expansion scheme to generate a large number of new samples to alleviate the problem of insufficient samples. On this basis, we introduced a novel hyperspectral classification network. The network first utilizes a module based on CNN-Transformer to extract blocks of hyperspectral images, where CNN focuses primarily on local features, while the Transformer module focuses mainly on non-local features. Furthermore, a simple channel-spatial attention module is adopted to further optimize the features. We conducted experiments on multiple hyperspectral image datasets, and the experiments verified the effectiveness of our CTA-net.

1. Introduction

With the advancement of remote sensing imaging technology, Hyperspectral Imaging (HSI) has become increasingly accessible. Unlike ordinary natural images that only contain three channels, hyperspectral remote sensing images cover a much broader spectral range, including the visible-light, near-infrared, mid-infrared, and far-infrared regions, typically with dozens or even hundreds of continuous spectral bands (Blanes, Serra-Sagristà, Marcellin, & Bartrina-Rapesta, 2012). Based on this rich and detailed spectral information, hyperspectral images can usually identify fine categories of objects, and the identification of these fine categories is very important for Earth observation (Gu, Chanussot, Jia, & Benediktsson, 2017). In recent years, HSI has been widely applied in various remote sensing applications, including urban planning (Huang & Zhang, 2008), resource exploration (Zhang et al., 2023), environmental monitoring (Ma et al., 2021), smart agriculture (Gevaert, Suomalainen, Tang, & Kooistra, 2015), military security (Tiwari, Arora, & Singh, 2011), etc. Among these applications, hyperspectral classification technology is one of the fundamental and

extremely important technologies. In recent years, the spectral and spatial resolution of hyperspectral images has been continually improving. These high-resolution images give much more information about the land covers. Nonetheless, the considerable spectral redundancy and the extremely high data dimensionality characteristic of hyperspectral image data pose substantial obstacles in the process of hyperspectral classification.

The curse of dimensionality often leads to poor performance in hyperspectral image classification (Gu et al., 2017; Lunga, Prasad, Crawford, & Ersoy, 2014). To address the challenges posed by the high dimensionality of hyperspectral data, many researchers in the early days were dedicated to developing a series of dimensionality reduction algorithms to explore the intrinsic structure of hyperspectral data in low-dimensional spaces, making the data easier to classify and achieving better classification performance (Lunga et al., 2014). For instance, algorithms based on Principal Component Analysis (PCA) (Du & Fowler, 2008; Kang, Xiang, Li, & Benediktsson, 2017) and Linear Discriminant Analysis (LDA) (Du, 2007; Jia et al., 2021; Shahdoosti &

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Mirzapour, 2017) were developed. The goal of these techniques is to preserve the intrinsic structure of the hyperspectral image data or its low-dimensional embedding relationships in the reduced-dimensional space.

However, because most of these traditional hyperspectral image classification techniques and models are crafted manually, their performance can be constrained, potentially impacting the precision of hyperspectral classification. This is particularly true in scenarios where labeled samples are sparse, leading to a noticeable drop in the classification accuracy of these conventional algorithms.

In recent years, since deep learning has achieved a series of milestone results in many image processing tasks, researchers have also introduced deep learning into various fields of remote sensing image processing, such as change detection (Dong, Yang, Qu, Xiao, & Li, 2023; Hu, Wu, & Zhang, 2022; Luo et al., 2023), anomaly detection (Gao et al., 2023; Zhu, Du, Dong, & Zhang, 2022), unmixing (Liu, Wang, Du, & Qu, 2023; Qu, Pan, Xu, Li, & Shi, 2023), image classification (Duan et al., 2024; Li, Li, Zhu, Liu, & Jiao, 2019; Lin, Fu, Wang, Xu, & Sun, 2017), and target detection (Chen et al., 2023; Guo, Yang, Wang, & Gao, 2021; Guo, Yang, Wang, Song, & Gao, 2020; Xie, Zhang, Li, Wang, & Du, 2020). All of these deep learning-based algorithms have achieved good results. In the field of hyperspectral image classification, researchers have also developed a series of deep learning-based algorithms (Shi & Pun, 2020). These models include Stacked Autoencoders (SAEs) (Shi & Pun, 2020), Recurrent Neural Networks (RNNs) (Hang, Liu, Hong, & Ghamisi, 2019; Mou, Ghamisi, & Zhu, 2017), Bayesian methods (Haut, Paoletti, Plaza, Li, & Plaza, 2018; Joshaghani, Davari, Hatamian, Maier, & Riess, 2023), Graph Convolutional Networks (GCNs) (Bai et al., 2022; Dong, Liu, Du, & Zhang, 2022), and Transformers (Fang, Ye, Sun, Zheng, & Wu, 2023; Hong et al., 2022; Shi et al., 2025; Wang, Zhang, Du, Zhang, & Tao, 2023; Yang, Du, & Wu, 2022). Compared with traditional hyperspectral algorithms, the aforementioned algorithms can usually perform end-to-end learning, better capturing the detailed features of hyperspectral images, thereby achieving higher classification accuracy. On this basis, algorithms based on multiscale structures, 3D-CNN (Sun, Zheng, Lu, & Wu, 2020; Zhong, Li, Luo, & Chapman, 2018), parallel dual-branch networks (Zhang et al., 2022), and attention-based networks (Roy, Manna, Song, & Bruzzone, 2021; Zhu, Jiao, Liu, Yang, & Wang, 2021) have also been proposed, and these algorithms have achieved even much better classification results. In addition, in recent years, some algorithms based on new architectures, such as mamba (Li, Luo, Zhang, Wang, & Du, 2024; Yang, Li et al., 2024; Zhou, Kamata, Wang, Wong, & Cynthia Hou, 2025) and KAN (Jamali, Roy, Hong, Lu, & Ghamisi, 2024), have also made some progress, providing some new ideas for the design of hyperspectral classification networks. Besides the innovation of network architecture, in the literature (Yang, Du, Wang and Zhang, 2024), in order to relieve the difficulty of pixel-level sample annotation, Yang et al. first proposed an innovative hyperspectral image classification method based on weak annotation, which can perform pixel-level classification by using image-level weak labels. It is an extremely promising algorithm.

The aforementioned deep learning-based hyperspectral image classification algorithms have made significant strides in improving the hyperspectral classification accuracy through adaptive end-to-end learning. However, these algorithms still face the challenge of achieving high classification accuracy when dealing with a limited number of labeled hyperspectral samples. In practical hyperspectral classification applications, although hyperspectral image data have become extremely easy to acquire, obtaining a large number of representative and accurately labeled hyperspectral samples remains a formidable task. Accurate labeling often requires field surveys and other reference information, which is difficult to obtain in many cases. For instance, in extremely remote areas such as uninhabited high-sea regions or high-altitude areas, acquiring a large number of labeled samples for hyperspectral

image resource exploration is nearly impossible. Consequently, the issue of insufficient labeled training sample size persists in hyperspectral classification.

In hyperspectral classification, the scarcity of labeled samples poses a significant challenge to accurate classification. On the one hand, when designing hyperspectral image classification networks, complex networks may struggle to learn generalized features due to the insufficiency of training samples, leading to overfitting on the training data and poor classification performance on test data. On the other hand, due to the scarcity of samples, mining more representative local and non-local hyperspectral features from fewer samples is an extremely difficult problem. Therefore, hyperspectral image classification with a limited number of labeled samples still faces significant challenges.

Some previously algorithms discussed the issue of having inadequate labeled samples for hyperspectral image classification. However, many of these algorithms require additional datasets for pre-training or rely heavily on a large volume of unlabeled data for self-supervised learning, which is subsequently fine-tuned using labeled samples (Li, Guo et al., 2023; Xu, Du, & Zhang, 2024). In addition, in real-world applications, it is also challenging to acquire additional data or a significant amount of unlabeled high-quality data.

In this paper, we propose a novel hyperspectral image classification algorithm, CNN-Transformer and Attention-based net(CTA-net) for hyperspectral image focus on the few-sample scenarios. Firstly, we design a new sample amplification algorithm for hyperspectral image which can generate a diverse set of synthetic samples from the limited labeled data. The proposed sample simplification method utilizes a generative approach based on the statistical characteristics of the hyperspectral data, which can relieve the burden of limited labeled samples. Based on the sample simplification module, we then introduce a new deep neural network for hyperspectral image classification. In the network, to fully explore the local and non-local information on patches of hyperspectral images, we introduce a new hybrid module based on CNN and Transformer (CT) block. This module capitalizes on the powerful local information representation capability of CNNs and the non-local information representation capability of Transformers. On this basis, we propose a novel attention module to further enhance the nonlinear representation capability of the network. We conduct experiments on several datasets and the results show the effectiveness of the proposed algorithm. The main contributions of this paper are as follows.

1. We introduce a new sample amplification algorithm based on the mixed strategy for hyperspectral image classification, which can relieve the burden of limited labeled samples.
2. We propose a hybrid module based on CNN and Transformer, CT block, to explore local and non-local information in hyperspectral images. In this module, the CNN is utilized to mine local information, while the Transformer is employed to extract non-local information, combining both types of information to achieve a better representation.
3. Building upon the CNN-Transformer hybrid module, to enhance the network's nonlinear representation ability, we have employed a novel attention module, which deepens the network while strengthening its nonlinear representation capabilities.
4. Based on the three modules mentioned above, we have proposed a novel hyperspectral image classification algorithm called CTA-net. We conducted experiments on several commonly used hyperspectral images, and the experiments have verified the effectiveness of the proposed algorithm.

The rest of the paper is as follows. In Section 2, we provide a detailed description of the proposed hyperspectral classification algorithm, CTA-net. Section 3 is the experimental section, where we conducted experiments on several hyperspectral image datasets and compared them with some state-of-the-art hyperspectral image classification algorithms. The Section 4 presents the conclusions of this paper.

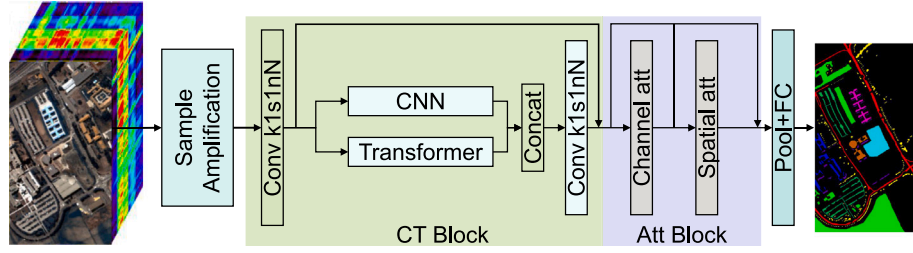


Fig. 1. Overall framework of CTA-net. The proposed CTA-net contains four main parts, including a sample amplification block, a CT block, an Att block(Channel-Spatial attention) and a simple classifier.

2. Proposed method

2.1. Overview

The overall framework of the CTA algorithm that we propose is depicted in Fig. 1. This framework is primarily composed of four components: a sample amplification module, a hybrid feature extraction module based on the CT module, and a feature enhancement module based on the attention module. The sample amplification module attempts to generate series of new samples based on the labeled sample to alleviate the problem of insufficient samples. Based on this, a CNN-Transformer mixture block, CT block, is used to explore the shallow features, while the CNN block is used to explore the local features and the Transformer block is used to explore the non-local features. In order to further optimize the extracted features by combining the inherent properties of the hyperspectral image blocks, CTA-net introduced a simple channel-spatial attention module. Finally, a global average pooling and a fully connected layer constitute the final classifier.

The hyperspectral image is input into the system, first passing through a convolutional layer that transforms the feature channels into n , then it is fed into the local and non-local feature extraction module. After that, it goes through a new attention module to optimize the features, and finally, it is entered into the classifier to obtain the final results. The subsequent sections will elaborate on the algorithmic modules presented in this paper.

2.2. Sample amplification module

When the training samples are extremely limited, training neural networks often face the problem of underfitting. At this time, existing training samples can be used and sample augmentation techniques can be utilized to generate a large number of labeled new samples for the network training, thus alleviating the problem of insufficient sample size.

The commonly used sample-augmentation techniques in traditional natural image classification algorithms include mainly translation, flipping, scaling, cropping, color transformation, and adding noise. However, in the field of hyperspectral image classification based on patches, the importance of the central pixel of a hyperspectral patch is usually much greater than that of the surrounding pixels. Moreover, due to the small spatial size of the patch and the extremely large number of channels, most of the sample augmentation techniques used in natural image classification algorithms cannot be directly applied to the sample augmentation of hyperspectral image patches.

In the sample augmentation of hyperspectral image patch samples, the center pixel is much more important than the other pixels. In order to ensure that the generated samples correspond to their original labels, we should try our best to ensure that the central pixel and its surrounding pixels of the generated hyperspectral patch are as similar or identical as possible to the original hyperspectral image block. In this paper, we use three simple augmentation strategies to expand the number of hyperspectral samples. One is to add random noise to the pixels outside the central region. Another is to perform random

rotation on the image block and the last is to generate new patch samples by linear summation of the same-class samples. The three sample augmentation techniques can be simply represented as follows:

$$\begin{aligned}\hat{X}_1 &= Addnoise(X) \\ \hat{X}_2 &= Rotation(X) \\ \hat{X}_3 &= Linear(X) \\ \hat{X} &= \{\hat{X}_1, \hat{X}_2, \hat{X}_3, X\}\end{aligned}\quad (1)$$

where X is the original hyperspectral image patches and $\hat{X}_k$ are the generated samples and the final dataset we used is $\hat{X}$. *Addnoise* means adding random Gaussian noise, *Rotation* means Random rotation ± 180 degrees, *Linear* means generating new patch samples by linear summation in some samples of the same class. To achieve linear summation, we choose a random sample in one class and sum it to other samples in the same class. In the process of increasing the sample size, we increase the sample class by class. In order to ensure that the training process includes the original samples, our augmented sample set not only contains samples generated by augmentation techniques but also includes samples of the original categories. It is worth noting that in the rotation and adding Gaussian noise sections, all samples participated in the sample augmentation process.

2.3. CNN-Transformer based module

In the past decade of computer vision research, a multitude of studies have demonstrated the powerful capability of convolutional neural networks (CNNs) to represent local information. The advancements of CNNs in the field of remote sensing image processing and hyperspectral image analysis have also shown their effectiveness. However, for hyperspectral images, nonlocal information is as crucial as local information when it comes to image classification. Taking full advantage of local and non-local information and integrating it is key to achieving efficient classification. In recent years, Transformer has proven its strong capability in extracting non-local information. In order to simultaneously mine local and non-local information in hyperspectral image patches, we propose a hybrid feature extraction module based on CNN and Transformer, CT block, whose main structure is shown in Fig. 1. We first input the original image into a convolutional layer for feature alignment, and then use CNN and Transformer modules, respectively, to extract local and non-local information. Afterwards, the local and non-local features extracted by the two networks will be concatenated together, and then a convolutional layer will be used to reduce the number of channels to the original number of channels and perform a residual connection with the input features.

The CNN module and the Transformer module that we used are shown in Fig. 2. As shown in the figure, for the CNN module, we use four different convolutional branches to extract features, then fuse them together, and use 2D convolutions with convolution kernel sizes of 1×1 to reduce the channel dimension to the same as the input and perform a residual connection with the input. In the four convolution branches, we used 2D convolutions with convolution kernel sizes of 1×1 and 3×3 , as well as multiple stacks of 2D convolutions with

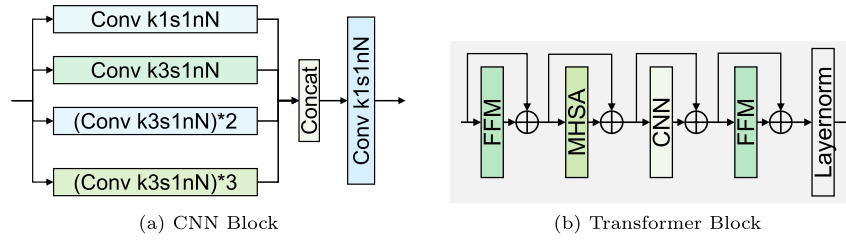


Fig. 2. The CNN block used in the CT block, where $Conv k1s1nN$ indicates a 2D convolution with a convolution kernel size of K , a stride of S , and an output channel number of N .

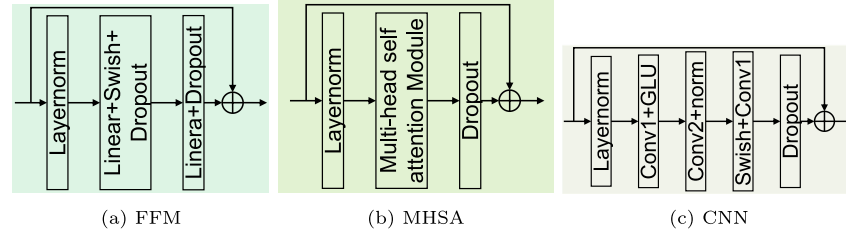


Fig. 3. The module of the Transformer block used in the proposed algorithm. The Transformer used in the proposed algorithm is the Conformer (Peng et al., 2021). In the Transformer block, FFM, MHSA and CNN block are shown in subfigures.

convolution kernel sizes of 3×3 to improve the receptive field of the two branches. In this way, we realize the aggregation of local information at different scales and improve the non-linear representative ability of the CNN module. It is worth mentioning that between the multiple 2D convolution layers stacked with convolution kernel sizes of 3×3 , we used a Batchnorm layer and a GELU activation layer and N is the channel of output features.

The Transformer module we use is derived from Conformer (Peng et al., 2021), and its structure is shown in the right figure of Fig. 2. As shown in the figure, the conformer module mainly includes two FFM modules, one MHSA module, and one CNN module, and each module uses residual connections. The FFM, MHSA and CNN modules are shown in Fig. 3. The various diagrams of the module are shown in the figure, where the multi-head self-attention module in the MHSA module uses relative position encoding. The CNN in Conformer is used mainly to obtain the partial receptive field, where conv1 represents the 2D convolution with a kernel size of 1×1 , and conv2 represents the 2D convolution with a kernel size of 3×3 .

2.4. Channel-spatial attention

In the case of limited training samples, designing overly complex networks may lead to an inability to properly learn the essential features of the data, resulting in underfitting or overfitting. However, using only a single CT block is not sufficient to fully exploit the channel-spatial features in the shallow layers of hyperspectral image blocks. For hyperspectral image classification, both spatial and spectral features are extremely important. To further optimize the learned features, we propose a simple Channel-Spatial Attention Network, whose architecture is shown in Fig. 4.

Our network architecture consists of two parts: the channel attention module and the spatial attention module. In the channel attention module, we learn a weighted value from each channel of the input feature, which is then multiplied by each channel of the image, followed by a residual connection with the input. In the reshape contain 1D convulsion layers with a kernel size of 3. In the spatial attention module, the input features are processed to obtain the most important features in space. The spatial information used in spatial attention contains several spatial statistics features, including the maximum, minimum, mean, and standard deviation. Once these values have been extracted, the information will be preliminarily extracted using 2D

convolution with a kernel size of 5×5 , as well as a PReLU activation function. Then, the features of the two branches are concatenated together, and a convolution with a kernel size of 1×1 is used to reduce the dimensionality of the channel. After that, it is multiplied pixel by pixel with the input features to obtain a weighted result. Finally, a residual connection is made with the input features.

3. Experiments

In this section, we detail the experiments. Firstly, we outline our experimental setup, which includes the hyperspectral image datasets utilized, evaluation metrics, comparison algorithms, and implementation specifics. Subsequently, we performed an in-depth analysis and comparison with several advanced hyperspectral image classification methods from both qualitative and quantitative perspectives. Lastly, to confirm the efficacy of our proposed algorithm, we performed thorough ablation studies.

3.1. Experimental setup

3.1.1. Datasets

In this paper, we conducted experiments on three hyperspectral image datasets to validate the effectiveness of our algorithm. The dataset used in our experiment includes the University of Pavia (UP) and WHU-Hi-HongHu (Honghu) and WHU-Hi—HanChuan (Hanchuan). Their specific information is shown below:

Pavia University

The UP dataset was acquired during a flight campaign over Pavia University in Northern Italy in 2001 using the Reflective Optics System Imaging Spectrometer (ROSIS). Some bands contain a large amount of noise, and after eliminating 12 noisy bands, this dataset we used only contains 103 bands measuring 610×340 . The UP dataset has a spatial resolution of 1.3 m per pixel and a wavelength range of 0.43–0.86 μm . This dataset contains a total of 42,776 labeled pixels with 9 different urban land cover categories. The detailed information of the UP dataset used in our experiments is shown in the Table 1.

WHU-Hi-HongHu

The Honghu dataset is a hyperspectral image with both high spatial and spectral resolution, which is captured on November 20, 2017, in Honghu City, Hubei province, China. This dataset was obtained

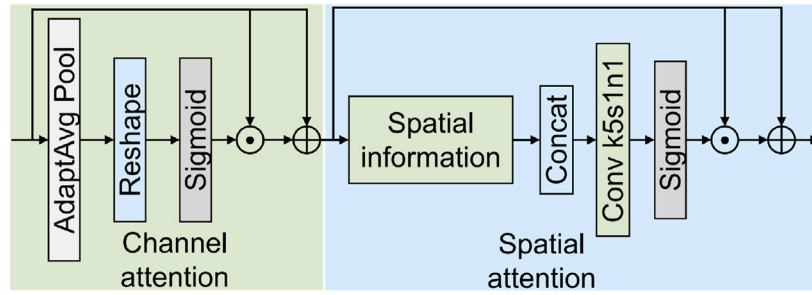


Fig. 4. The channel-spatial attention module proposed in this paper.

Table 1

Category information of the Pavia university data set.

Class	Color	Class name	Training	Validation	Test
1		Asphalt	10	5	6616
2		Meadows	10	5	18 634
3		Gravel	10	5	2084
4		Trees	10	5	3049
5		Painted-m-s	10	5	1330
6		Bare-soil	10	5	5014
7		Bitumen	10	5	1315
8		Self-block-b	10	5	3667
9		Shadows	10	5	932
Total			90	45	42 641

Table 2

Category information of the WHU-Hi-HongHu data set.

Class	Color	Class name	Training	Validation	Test
1		Red roof	10	5	14 036
2		Road	10	5	3497
3		Bare soil	10	5	21 806
4		Cotton	10	5	163 270
5		Cotton firewood	10	5	6203
6		Rape	10	5	44 542
7		Chinese cabbage	10	5	24 088
8		Pakchoi	10	5	4039
9		Cabbage	10	5	10 804
10		Tuber mustard	10	5	1279
11		Brassica parachinensis	10	5	11 000
12		Brassica chinensis	10	5	8939
13		Small Brassica chinensis	10	5	22 492
14		Lactuca sativa	10	5	7341
15		Celtuce	10	5	987
16		Film covered lettuce	10	5	7247
17		Romaine lettuce	10	5	2995
18		Carrot	10	5	3202
19		White radish	10	5	8697
20		Garlic sprout	10	5	3471
21		Broad bean	10	5	1313
22		Tree	10	5	4025
Total			220	110	384 119

using a 17 mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro UAV platform. The dataset is a high resolution hyperspectral remote sensing image dataset collected by unmanned aerial remote sensing, which features a spatial resolution of approximately 0.043 m and has dimensions of 940×475 pixels. This dataset contains 270 bands that cover a spectral range from 0.4 to 1 μm and includes 22 distinct land cover categories. Detailed information on the Honghu dataset used in our experiments is presented in Table 2.

WHU-Hi-HanChuan

The WHU-Hi-HanChuan dataset was obtained on the afternoon of June 17, 2016, in Hanchuan City, Hubei Province, China. This image was acquired with the same imaging equipment as Honghu, named the Headwall Nano-Hyperspec imaging sensor installed on the Leica Aibot X6 UAV V1 platform. The research area is a fringe zone of rural and

Table 3

Category information of the WHU-Hi-HanChuan data set.

Class	Color	Class name	Training	Validation	Test
1		Strawberry	10	5	44 720
2		Cowpea	10	5	22 738
3		Soybean	10	5	10 272
4		Sorghum	10	5	5338
5		Water spinach	10	5	1185
6		Watermelon	10	5	4518
7		Greens	10	5	5888
8		Trees	10	5	17 963
9		Grass	10	5	9454
10		Red roof	10	5	10 501
11		Gray roof	10	5	16 806
12		Plastic	10	5	3664
13		Bare soil	10	5	9101
14		Road	10	5	18 545
15		Bright object	10	5	1121
16		Water	10	5	75 386
Total			160	80	275 290

urban areas, which contains buildings, water bodies, and cultivated land, with seven species of crops and other land covers. The image size of this dataset was 1217×303 pixels, and there were 274 bands of 400 to 1000 nm, and the spatial resolution of the hyperspectral image carried by drone was approximately 0.109 m. It is worth noting that because the WHU-Hi-HanChuan dataset was acquired in the afternoon when the solar elevation angle was low, there are many areas covered by shadows in the image. Detailed information on this dataset is shown in Table 3.

3.2. Experimental setting

According to Fu, Du, and Zhang (2024a) and Li, Liu, Xue, Huang and Yang (2023), we adopt several algorithms as compared algorithms. Comparison algorithms cover popular approaches to hyperspectral image classification, including ContextNet (Lee & Kwon, 2017) SSAN (Sun et al., 2020), RSSAN (Zhu et al., 2021), SSAtt (Hang, Li, Liu, Ghamisi, & Bhattacharyya, 2021), A2S2K (Roy et al., 2021), SSTN (Zhong, Li, Ma, Li, & Zheng, 2022), SSSAN (Zhang et al., 2022), CVSSN (Li, Liu et al., 2023), ReSCnet (Fu, Du, & Zhang, 2024b), and ETF (Fu et al., 2024a). To evaluate the classification performance of different algorithms and analyze the impact of various parameters, we performed quantitative and subjective evaluations. For the quantitative assessment, we adopt three evaluation metrics commonly used in HSI classification, including overall accuracy (OA), average accuracy (AA), Kappa coefficient (Kappa), and the accuracy of each categories; all metrics are evaluated in the test set.

In our experimental setup, a random selection of ten samples is utilized for training to mitigate bias. Each experiment is carried out ten times and the mean value of the results is considered to represent the final outcome. An improved performance of the model is indicated by higher average values of these metrics. In addition to

Table 4

Quantitative classification results of the UP dataset with 10 labeled training samples per class. Best results are shown in bold.

Class	ContextNet	RSSAN	SSTN	SSAN	SSSAN	SSAtt	A2Snet	CVSSN	ReSCnet	ETF	Proposed
1	93.25	81.43	94.88	83.05	94.63	94.63	96.69	97.65	95.93	96.13	98.27
2	88.25	85.15	97.17	94.32	88.45	95.51	97.26	96.82	99.53	97.24	99.32
3	49.02	35.70	83.35	52.48	57.30	60.97	68.37	87.07	89.75	91.70	93.99
4	71.60	80.16	72.86	73.35	86.90	87.64	81.58	92.45	74.91	78.34	81.26
5	97.19	72.91	93.31	97.48	94.91	97.36	95.71	99.78	95.22	95.26	96.66
6	48.60	49.22	72.48	55.59	53.05	63.87	73.42	76.55	82.60	79.12	88.59
7	49.91	34.96	81.94	41.22	58.15	55.48	68.84	77.83	97.16	85.56	96.96
8	78.49	65.96	81.99	74.02	80.96	79.85	88.47	86.45	89.76	88.41	90.24
9	66.22	60.74	89.14	84.45	97.56	88.63	92.72	95.33	93.00	87.35	90.28
OA	72.71	68.90	86.10	75.40	77.65	82.19	87.02	90.66	92.09	90.05	94.42
AA	71.39	62.91	85.12	72.88	79.10	80.43	84.79	89.99	90.87	88.79	92.84
Kapppa	65.40	59.31	82.23	68.88	71.08	77.21	83.30	87.83	89.74	87.09	92.71

quantitative evaluation, subjective visual assessments are performed by generating and examining the classification maps for each algorithm against the ground truth. This methodology enables a straightforward visual comparison of the efficacy and performance of the algorithms. A classification map that closely aligns with the ground truth is indicative of superior performance.

The experiment was conducted on a system utilizing an Intel(R) Xeon(R) Silver 4310 CPU operating at 2.10 GHz along with a NVIDIA GeForce RTX 3090 GPU, featuring 24 GB of memory for graphics processing. The operating system was Ubuntu. The algorithm in question was developed in Python 3.6 with the aid of the PyTorch 1.10 library. Throughout the training phase, Adam Optimizer was used with a steady learning rate of 0.00008, and a batch size of 32 was kept uniform across all datasets. Each trial consisted of 150 epochs to confirm convergence. To ensure unbiased comparison, every algorithm was subjected to ten trials starting from random initial conditions, and the mean and variance of the results were recorded as ultimate performance metrics. The spatial window size of the 3D patch was set at 15×15 (although ReSCnet and ETF employed 11×11 , which can achieve the best performance).

3.3. Results analysis

3.3.1. Quantitative accuracy analysis

The quantitative results are shown in Tables 4, 5, and 6. In the three tables, we compare our algorithm with other algorithms compared in terms of classification performance in three datasets using metrics such as the average classification accuracy of each class, OA, AA, and Kappa coefficients.

The experimental results for the UP dataset are shown in Table 4. As shown in the table, apart from the algorithm we proposed, ETF and ReSCnet have achieved the best results among all comparison algorithms. These two algorithms are designed for small samples. In the ETF algorithm, the ETF classifier first pre-allocates a category vector, and then uses this vector and the tensor learned from the sample to ensure aggregation within the same category and separation between different categories, thereby achieving good performance. In ReSCnet, lightweight modules are more convenient for optimization. In addition to these two algorithms, CVSSN, which integrates spatial information with spectral information, has also achieved very good classification performance. From the content in the table, we can further observe that in the UP dataset, the proposed algorithm has achieved the best results in the classification accuracy of most categories of ground objects. Compared with ReSCnet, the proposed algorithm has increased by 2.33, 1.97 and 2.95 percentage points in OA, AA and Kappa, respectively.

The visualization results of the classification map on the UP dataset are shown in Fig. 5. From the classification map, we can also draw similar conclusions. Among all algorithms, the classification error of the proposed algorithm is less than that of other algorithms. In addition to the proposed algorithm, algorithms such as CVSSN, ReSCnet, ETF, and SSSAN have also achieved good classification results.

The classification results on the Honghu dataset are shown in Table 5. As shown in Table 5, similar conclusions can also be drawn on the Honghu dataset as on the UP dataset. The difference is that the Honghu dataset has more classification categories, and there is a certain imbalance between the categories, making the classification more difficult compared to the UP dataset. Among all comparison algorithms, the ReSCnet algorithm still achieved the best results. Compared to ReSCnet, the proposed algorithm is 2.23, 3.21, and 2.75 percentage points higher in OA, AA, and Kappa, respectively. It is worth mentioning that due to the class imbalance on the Honghu dataset, the OA of most algorithms on this dataset is higher than the AA. Compared with all other algorithms, the proposed algorithm has the highest AA. In most categories, the proposed algorithm has achieved the highest classification accuracy. The visualization results of the classification map on the Honghu dataset are shown in Fig. 6. From the figure, it can be seen that compared to other comparison algorithms, the classification accuracy of the proposed algorithm is relatively lower.

The classification results on the Hanchuan dataset are shown in Table 6. The Hanchuan dataset is similar to the Honghu dataset, both containing a relatively large number of categories. There is also a certain degree of class imbalance in this dataset, therefore, the OA accuracy value on this dataset is greater than the AA accuracy. On this dataset, the ETF algorithm achieved the best classification results. In addition to ETF, ReSCnet also achieved competitive classification performance. Compared with other comparison algorithms, the proposed algorithm achieved the best classification accuracy. Compared to the RTF algorithm, the proposed algorithm improved by 3.74, 3.46, and 4.30 percentage points in OA, AA and Kappa, respectively.

The visualization of the classification map for the Hanchuan dataset is shown in Fig. 7. From the image, it can be seen that ContextNet, RSSAN, SSAN, and A2Snet contain a significant amount of classification noise, especially within the tiny fields on the right side, where the noise is particularly evident. In contrast, the classification maps of CVSSN, ReSCnet, ETF, and the proposed algorithm contain fewer classification errors. Among all the algorithms' classification maps, the proposed algorithm has a smaller difference from the GT.

3.4. Ablation study

To further validate the effectiveness of all of our proposed modules, we performed ablation experiments on three datasets while keeping the experimental parameter settings consistent. The results of the ablation experiments are presented in Table 7. In this table, "SA" refers to the Sample Augmentation module, "CT" refers to the CNN-Transformer-based module we proposed, and "CSA" refers to the Channel-Spatial attention module we introduced.

In the ablation experiment of the CT residual module, we excluded the Transformer module within this module, while keeping the other parts unchanged. For the ablation experiments involving the two outer modules, the respective modules were completely removed. In the table, a checkmark ✓ indicates the inclusion of its corresponding module.

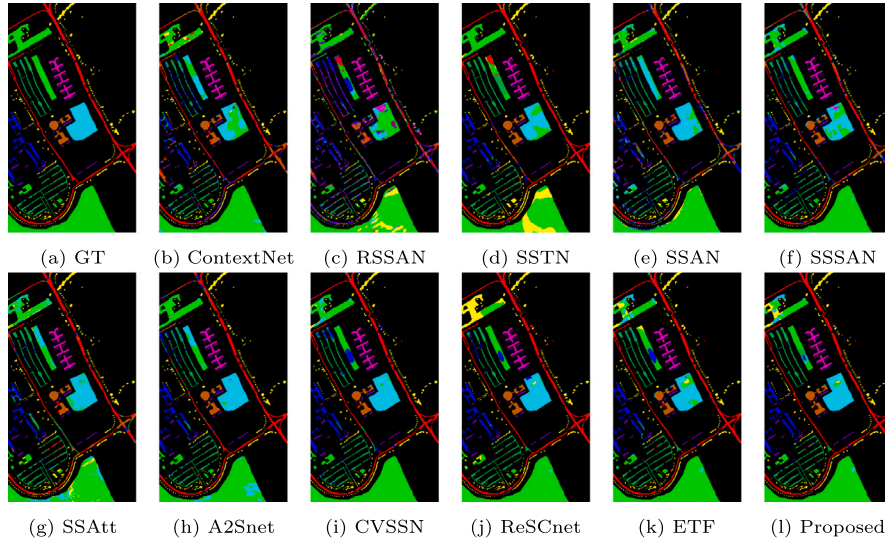


Fig. 5. The classification maps of UP.

Table 5

Quantitative classification results of the Honghu dataset with 10 labeled training samples per class. Best results are shown in bold.

Class	ContextNet	RSSAN	SSTN	SSAN	SSSAN	SSAtt	A2Snet	CVSSN	ReSCnet	ETF	Proposed
1	89.00	88.36	98.48	94.68	94.98	96.13	92.31	96.01	99.21	98.92	99.05
2	65.16	42.39	63.86	63.60	66.45	69.48	63.17	68.17	57.80	61.50	60.59
3	89.04	82.26	96.35	92.30	90.12	92.74	91.99	96.14	96.82	95.54	96.53
4	98.60	98.41	99.46	99.07	99.21	99.00	99.31	98.76	99.75	99.55	99.66
5	20.88	17.12	41.57	35.63	31.00	33.70	34.01	52.48	62.00	52.22	63.84
6	90.04	85.70	97.06	93.94	88.60	95.06	92.08	94.91	97.86	97.50	96.45
7	79.31	61.99	92.10	84.05	76.35	87.74	81.00	85.69	90.69	92.85	94.50
8	21.09	12.77	35.63	33.26	24.74	27.79	27.78	52.47	46.38	57.48	66.16
9	91.23	83.05	97.49	93.65	92.83	96.87	93.43	97.92	98.71	97.41	98.05
10	59.50	42.97	88.52	75.73	73.10	70.76	80.84	87.75	86.94	92.08	94.62
11	56.02	29.09	70.25	47.88	51.27	66.15	57.72	73.51	80.90	83.54	91.05
12	47.27	37.40	66.24	62.68	48.37	63.48	59.64	61.80	66.55	70.02	71.10
13	68.68	64.10	74.31	76.42	70.10	75.08	75.55	85.03	85.95	88.17	90.90
14	49.49	38.68	91.27	76.85	66.80	74.54	73.59	87.14	92.46	92.84	94.19
15	37.98	14.20	72.41	46.63	36.58	55.04	18.89	72.55	76.30	81.64	75.80
16	89.99	63.69	98.03	92.01	87.98	93.21	92.02	95.21	97.53	96.99	96.89
17	48.69	30.22	80.53	71.71	46.07	60.69	68.57	83.22	93.84	91.57	90.34
18	35.56	22.14	55.57	50.79	44.14	48.61	49.61	59.80	70.51	73.17	76.82
19	75.30	49.71	79.25	78.16	74.86	83.42	72.49	87.16	91.06	88.69	93.71
20	64.50	32.52	65.29	62.00	55.40	58.92	54.64	66.34	78.60	75.15	80.15
21	18.94	7.93	28.88	13.49	16.47	14.44	25.30	19.63	36.48	40.34	48.11
22	31.68	18.58	56.44	29.49	29.35	44.09	37.57	44.46	60.65	49.04	58.83
OA	74.65	63.94	86.42	77.39	75.99	80.86	79.79	85.99	90.61	89.87	92.84
AA	60.36	46.38	74.95	67.00	62.03	68.50	65.52	75.73	80.31	80.69	83.52
Kappa	69.26	57.03	83.18	72.99	71.04	76.54	75.44	82.64	88.28	87.40	91.03

Please note that all ablation experiments are the average results of ten randomized experiments. The table presents the performance metrics for each configuration, allowing us to analyze the impact of each module on the overall classification performance.

From the table, it is clear that our three proposed indicators have considerably boosted the classification accuracy in all datasets. For instance, in the UP dataset, the OA of the modules increased to 0.37, 8.26, and 1.23, respectively, highlighting the efficacy of the modules. It should be noted that, across the three datasets, removing the CT module results in the greatest decrease in overall classification accuracy. This indicates that our CT module is the most crucial component of the proposed network. Building on the CT module, optimizing features with the attention module, and alleviating the issue of insufficient samples using sample enhancement techniques, the combination of these three elements achieves the best possible classification performance. Overall, the ablation experiments confirm that our modules markedly enhance the classification accuracy across datasets, supporting their effectiveness in the classification of hyperspectral images.

3.5. Partial parametric analysis

We further analyzed some important parameters of the proposed algorithm. These mainly include the number of samples used, the channel of the model, and the size of the hyperspectral image patches.

In terms of the number of samples for the train, we experimented with samples [5,10,15,20,25,30,40,50,100], and the classification accuracy of the three datasets is shown in Fig. 8. As can be seen in the figure, as the number of samples increases, the overall classification accuracy shows an upward trend. Even when the sample size is extremely small, the proposed algorithm can still achieve good classification performance, which shows that the proposed algorithm can achieve good classification results under different sample conditions.

The classification results using different numbers of channels are shown in Fig. 9. As illustrated, on the UP dataset, as the number of channels increases, the accuracy gradually improves. However, after the number of channels exceeds 128, further increasing the number of channels actually leads to a decrease in accuracy. In the other two

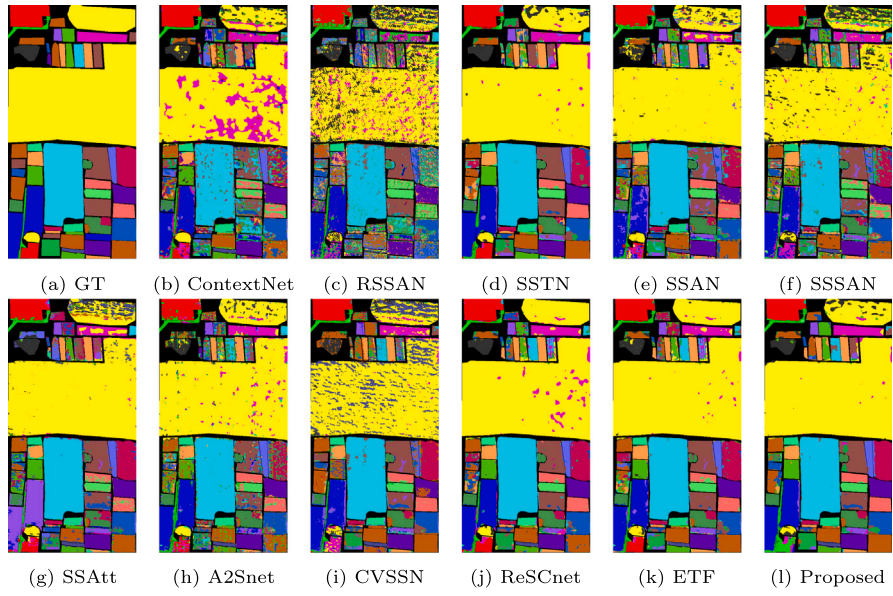


Fig. 6. The classification maps of Honghu.

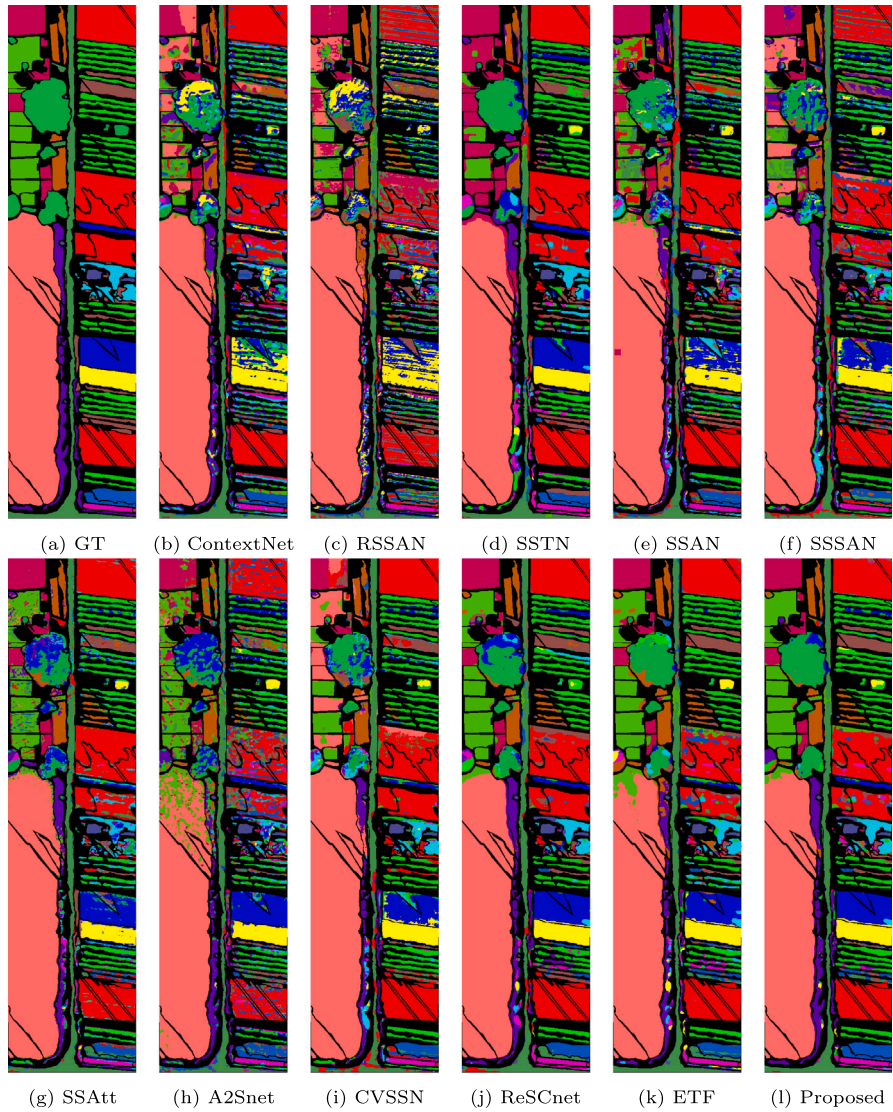


Fig. 7. The classification maps of Hanchuan.

Table 6

Quantitative classification results of the Hanchuan dataset with 10 labeled training samples per class. Best results are shown in bold.

Class	ContextNet	RSSAN	SSTN	SSAN	SSSAN	SSAtt	A2Snet	CVSSN	ReSCnet	ETF	Proposed
1	93.83	88.05	93.94	87.24	86.32	94.32	90.73	88.96	96.39	96.77	97.36
2	83.95	78.53	82.60	81.35	84.45	85.71	89.32	89.72	83.80	94.55	93.99
3	44.54	36.08	73.64	57.70	51.41	54.25	59.00	61.76	85.62	84.23	84.12
4	50.87	33.52	89.25	67.32	57.00	70.22	78.80	77.58	96.21	87.82	93.19
5	23.30	30.08	26.56	32.21	33.00	38.58	30.84	48.39	53.19	62.71	61.82
6	18.66	11.53	38.87	20.85	18.99	21.40	26.05	36.11	51.04	52.09	66.20
7	53.23	40.63	82.20	53.82	69.40	64.50	47.70	76.69	67.51	71.56	70.95
8	58.50	54.62	80.83	76.49	69.92	74.39	75.88	79.90	83.93	88.75	94.55
9	56.98	44.28	70.75	53.55	38.19	61.73	62.35	66.87	74.06	76.45	78.82
10	58.63	48.30	80.32	70.83	62.19	73.01	59.83	80.44	98.67	95.75	97.40
11	45.34	60.49	68.32	66.06	68.17	61.59	64.72	84.73	87.15	76.16	87.67
12	35.04	24.62	36.61	31.27	39.45	43.87	41.10	67.97	53.57	59.03	67.65
13	36.11	32.94	48.42	33.56	32.98	42.65	42.77	53.33	59.77	60.70	74.08
14	75.30	64.36	83.12	74.69	81.14	78.61	76.15	84.27	87.26	86.35	88.32
15	39.82	33.30	51.17	68.59	54.55	61.59	44.85	79.42	54.90	59.59	55.96
16	93.84	98.44	98.89	97.03	97.08	98.14	98.14	93.16	99.56	99.20	99.65
OA	67.00	61.20	78.95	68.84	67.45	74.66	74.58	79.48	87.11	87.23	90.97
AA	54.25	48.74	69.09	60.78	59.02	64.03	61.76	73.08	77.06	78.42	81.98
Kappa	62.01	55.81	75.68	64.16	62.90	70.79	70.71	76.03	85.00	85.18	89.48

Table 7

Comparisons on model efficiency of different methods on UP, Honghu and Hanchuan datasets.

Dataset	SA	CT	CSA	OA	AA	Kappa
UP	✓	✓	✓	94.42 ± 1.90	92.84 ± 1.69	92.71 ± 2.43
	✓	✓	–	94.07 ± 2.22	92.13 ± 2.03	92.27 ± 2.83
	✓	–	✓	86.16 ± 4.60	82.90 ± 2.57	82.30 ± 5.42
	–	✓	✓	93.19 ± 2.48	91.40 ± 2.42	91.13 ± 3.15
Honghu	✓	✓	✓	92.84 ± 1.24	83.52 ± 2.40	91.03 ± 1.53
	✓	✓	–	92.24 ± 1.86	82.76 ± 2.64	90.30 ± 2.27
	✓	–	✓	89.55 ± 1.56	78.30 ± 1.58	86.86 ± 1.09
	–	✓	✓	91.63 ± 1.31	82.10 ± 2.26	89.53 ± 1.60
Hanchuan	✓	✓	✓	90.97 ± 1.63	81.98 ± 2.83	89.48 ± 1.89
	✓	✓	–	90.52 ± 1.87	81.67 ± 2.70	88.96 ± 2.16
	✓	–	✓	85.10 ± 2.01	73.90 ± 2.05	82.72 ± 2.26
	–	✓	✓	88.89 ± 1.50	78.60 ± 2.20	87.06 ± 1.74

datasets, as the number of channels increases, the accuracy continues to improve, but the increase in accuracy is limited after the number of channels exceeds 128. Therefore, in our main experiments, the network uses 128 channels.

Patch size is also a relatively important parameter. We set the patch sizes to range from 7×7 to 23×23 and the results are shown in Fig. 10. As illustrated, in the UP data set, patch sizes of 13×13 and 15×15 achieved the best classification precision. As the patch size increases further, the accuracy actually decreases. However, in the other two datasets, the accuracy increases consistently with increasing patch size. The main reason for this may be that the UP dataset contains many smaller patches of land cover, so the use of larger patches may also introduce certain amounts of other noise information, thereby affecting the classification results. In contrast, the Honghu and Hanchuan datasets have larger land patches, so the larger the patch size, the higher the classification accuracy. However, larger patches require more computational power. After the patch size exceeds 15×15 , the improvement in accuracy is limited, so we used a patch size of 15×15 in our main experiments.

4. Conclusion

In this paper, we propose a hyperspectral image classification algorithm for small sample scenarios. The network utilizes a sample augmentation strategy to generate a large number of samples from existing ones, alleviating the issue of insufficient sample quantity. Furthermore, we propose a CT block for feature extraction and employ channel-spatial attention to further optimize the extracted features. Our ablation experiments demonstrate the effectiveness of each module.

At the same time, experimental results also show that the proposed algorithm achieves a good classification performance. However, in fact, we did not use a large number of unlabeled samples. In practical applications, the information contained in these numerous unlabeled samples can be further explored to enhance the final classification performance, which will be our future research work.

CRedit authorship contribution statement

Chuan Fu: Writing – original draft, Validation, Software, Methodology.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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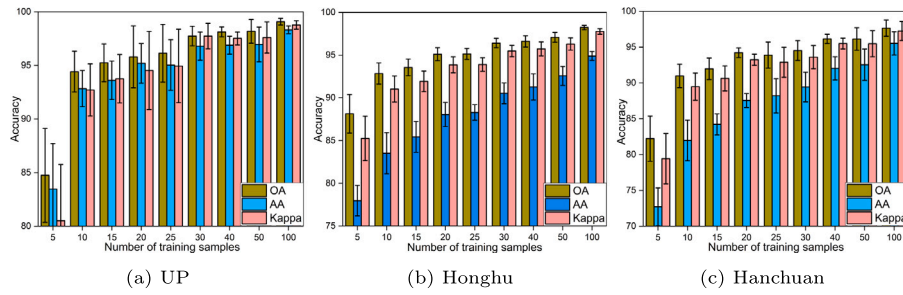


Fig. 8. The classification results of different number of training samples.

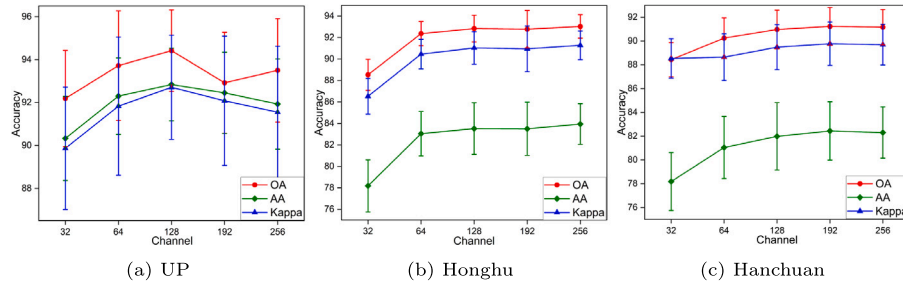


Fig. 9. The classification results of different number of channel.

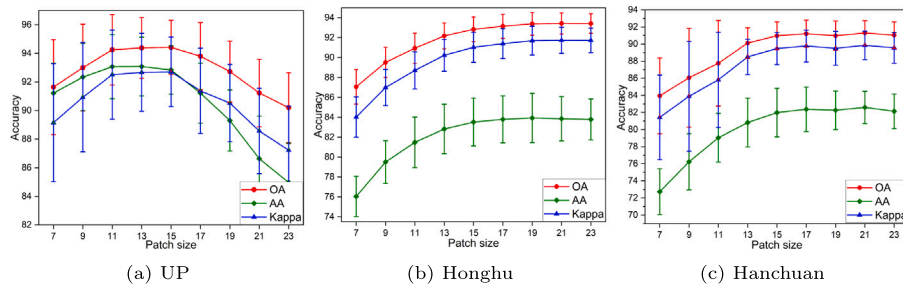


Fig. 10. The classification results of different patch sizes.

Data availability

No data was used for the research described in the article.

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